AUTOMOBILE ENGINEERING · PART 5 OF 12

Safety Engineering

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A car's safety has three jobs: help the driver **avoid** a crash, help everyone **survive** one, and help rescuers afterwards. Avoidance comes from brakes, tyres, ABS and stability control. Survival comes from a rigid passenger cell, crumple zones that absorb energy, seatbelts and airbags. In India, crash ratings from Bharat NCAP (government, since 2023) and Global NCAP (independent) now let buyers compare cars on tested performance instead of marketing. A star rating is meaningful only for the exact version tested and the protocol year it was tested under.

1. The physics: why speed is the enemy

In a crash, the car's kinetic energy (½ × mass × speed²) has to go somewhere. It goes into bending metal, which is the safety engineer's job to arrange. Doubling the speed quadruples the energy. Bharat NCAP's frontal offset test runs at 64 km/h; a crash at 120 km/h carries about 3.5 times that energy.* No structure can absorb that safely, which is why speed limits matter more than any feature.

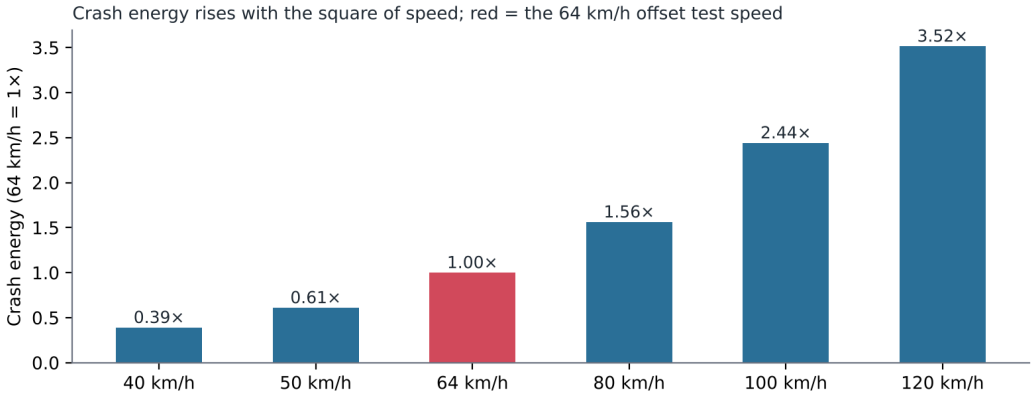


Fig 1. Relative crash energy against the 64 km/h test speed. Energy = ½ mv², so it scales with speed squared.

2. Three layers of safety

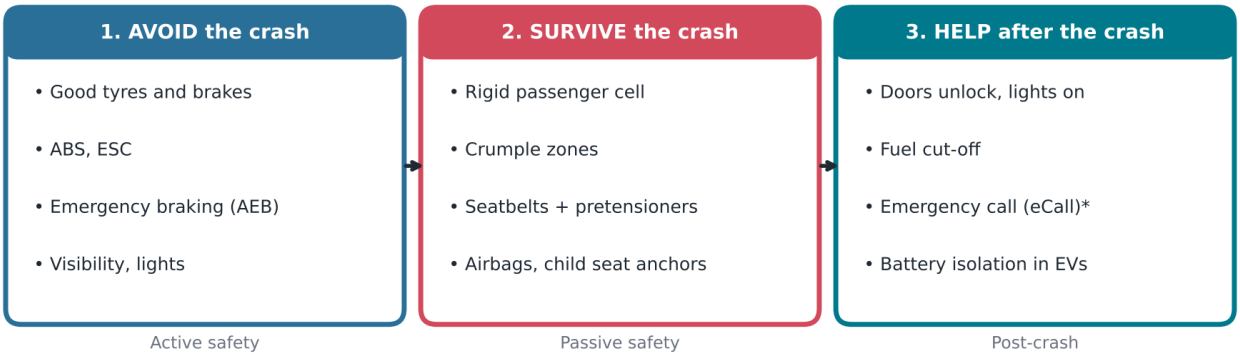


Fig 2. Safety as three layers: avoid, survive, help after the crash. (*eCall is not yet widely fitted in India.)

3. How a car survives a crash

- **Rigid passenger cell:** the cabin is built from very strong steel (often high-strength or hot-stamped grades) so it does not collapse onto occupants.
- **Crumple zones:** the front and rear are designed to fold progressively, absorbing energy over a longer time. Stopping over a longer time means lower force on the body.
- **Seatbelts with pretensioners and load limiters:** the belt tightens in the first milliseconds and then lets out a little so the chest is not injured. The belt is the single most important restraint; airbags assume you are wearing it.
- **Airbags:** a front airbag cushions the head and chest; side and curtain airbags protect in side impacts and rollovers. Six airbags are not mandatory in India: a January 2022 draft notification lapsed, and in September 2023 the transport minister said six airbags would not be made compulsory and that Bharat NCAP should push makers to fit them. Several popular models, including the Swift and Dzire, now fit six as standard.
- **Child restraints (ISOFIX):** standard anchor points let a child seat clip rigidly into the car. Child occupant protection is scored separately in NCAP tests.

4. How a car avoids a crash

- **ABS** prevents wheel lock under hard braking; **ESC** stops skids by braking single wheels. Global NCAP has pushed ESC as a key feature, and it is now rewarded in the protocol (see below).
- **Autonomous emergency braking (AEB)** uses a radar or camera to brake automatically when a collision is near. It is part of the driver-assistance systems in Part 8.
- **Tyres and brakes in good condition** matter more than any electronic feature; Part 4 covers stopping distances.

5. Crash tests: how to read the stars

Global NCAP, a UK-based charity, started crash-testing popular Indian cars in 2014 under its Safer Cars for India campaign. Its first round tested five cars, including the Swift, with no frontal airbags fitted; all scored zero stars for adult protection. Results pushed Toyota and Volkswagen to make frontal airbags standard across their Indian lineups.

Bharat NCAP is India's own programme. Its programme took effect on 1 October 2023 (announced 22 August 2023*), and crash tests began in December 2023. The current protocol (AIS-197) uses a 64 km/h frontal offset test, a 50 km/h side-barrier test and a pole side test, and carries ESC and pedestrian-protection requirements. The first 5-star results went to the Tata Harrier and Safari (adult 30.08/32, child 44.54/49). A tougher Bharat NCAP 2.0 (five crash tests, a 100-point scale, with curtain airbags and ESC required for any star rating) is in draft for October 2027. The programme is voluntary: makers choose which cars to submit.

Car and test	Protocol / year	Result
Maruti Dzire (4th gen)	Global NCAP, Nov 2024	Adult 31.24/34 (5 stars); child 39.20/49 (4 stars); first Maruti to score 5 stars
Maruti Dzire	Bharat NCAP	5 stars (29.46/41.57)
Maruti Swift (3rd gen)	Global NCAP 2.0, 2022	Adult 19.19/34 (1 star), child 16.68/49 (1 star); the older Swift scored 2 stars each in 2018 under the previous protocol
Tata Punch (petrol)	Global NCAP, 2021 (older 17-point scale)	Adult 16.45/17 (5 stars); child 40.89/49 (4 stars). Not comparable with 34-point scores
Tata Punch.ev	Bharat NCAP, 2024	Adult 31.46/32 (5 stars); child 45.00/49 (5 stars); the 2026 facelift scored 31.09/32 and 45/49

Car and test	Protocol / year	Result
Tata Nexon	Global NCAP and Bharat NCAP	5 stars in both (Bharat NCAP 29.41/43.83)
Tata Harrier, Safari	Bharat NCAP, Dec 2023	First Bharat NCAP results: adult 30.08/32, child 44.54/49 (5 stars each)
VW Taigun, Skoda Kushaq	Global NCAP, Oct 2022	Adult 29.64/34, child 42.00/49 (5 stars each)
Hyundai Creta	Global NCAP, 2022 (older protocol)	3 stars

Not every popular car has a Bharat NCAP rating yet: as of November 2026 I found no Bharat NCAP result for the Swift, Creta, Hyryder or City. The Hyundai Venue, Kia Seltos and Toyota Innova Hycross have all been reported at 5 stars in Bharat NCAP.

Why scores aren't directly comparable. Global NCAP changed its protocol in July 2022 (adding side-impact protection and rear seatbelt reminders, and making ESC and side head airbags requirements for higher ratings) and again in August 2025 (rewarding safety equipment only if fitted as standard). A 5-star result in 2021 and one in 2024 reflect different tests. Always check which protocol, which variant (airbag count differs by trim) and which year.

6. What it means for a buyer

- Choose the variant that matches the tested one: the rating usually applies to a specific trim and airbag count.
- Check which protocol and year the rating came from; newer protocols are stricter.
- Prefer cars with six airbags, ESC and ISOFIX as standard, not as paid extras.
- Wear the belt in the rear seat too; most rear passengers in India still do not.

Summary

- Safety has three layers: avoid the crash, survive it, help afterwards.
- Crash energy rises with speed squared, so a structure that is safe at 64 km/h cannot protect at 120 km/h.
- A rigid cell, crumple zones, belts with pretensioners and airbags protect occupants together.
- ABS, ESC and AEB help avoid the crash.
- Bharat NCAP (since 2023) and Global NCAP rate cars; ratings depend on the variant, the protocol and the year.
- Prefer six airbags, ESC and ISOFIX as standard.

*Unverified or general knowledge: the 3.5× comparison is arithmetic ($(120/64)^2$); the 22 August 2023 Bharat NCAP announcement date, eCall availability and the seatbelt-usage remark were not re-checked, and the Creta's 3-star Global NCAP result is from 2022 under an older protocol. Cross-checked in November 2026: six-airbag status (MoRTH draft of 14 January 2022; minister's statement of 13 September 2023), Bharat NCAP dates, protocol and results, Dzire, Punch, Punch.ev, Nexon, Harrier, Safari, Kushaq results, the Swift's 2022 Global NCAP result (1 star adult 19.19/34 and child 16.68/49; Autocar India) and 2018 result (2 stars), the Creta and i20 3-star result (Autocar Professional), and 5-star Bharat NCAP reports for the Venue, Seltos and Hycross (Rushlane, 91wheels, Gaadify). Protocol history: Global NCAP (via Wikipedia).

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 6.